QUANTUM COMPUTING • NOISE-AWARE ALGORITHM EVALUATION

NOISE ROBUSTNESS OF VQE AND QAOA

A comparative study using simulated depolarizing noise

QUISKIT SIMULATION **Four-Node Max-Cut** **Ideal Vs Noisy Conditions**



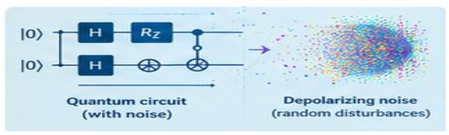
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EVENT -
Quantum Technology Meet · IIIT Delhi

01 INTRODUCTION

Quantum algorithms can be affected by noise in quantum circuits. This study uses Qiskit simulation to compare VQE and QAOA under selected depolarizing-noise strengths, examining average solution quality and run-to-run variability.



WHY NOISE MATTERS
Gate disturbances can change the measured state and affect the quality and consistency of an algorithm's output.

02 RESEARCH QUESTION

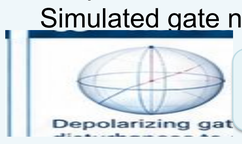
How does increasing quantum noise affect VQE and QAOA, and how do their robustness characteristics compare under the same simulation conditions?

- Compare ideal and noisy performance.
- Study selected depolarizing-noise strengths.
- Measure solution quality and variability.

IDEAL vs NOISY | SELECTED NOISE LEVELS | QUALITY + VARIABILITY

03 NOISE MODEL & SCOPE

Depolarizing gate noise models random disturbances to quantum states. The simulation applies selected noise strengths to circuit gates. The same settings are used for both algorithms.
Scope: a four-node Max-Cut instance; VQE with two rotation/entanglement layers; QAOA at p = 1.
Simulated gate noise only—not hardware measurements.



BENCHMARK SCOPE
4-node Max-Cut • VQE: two rotation/entanglement layers • QAOA: p = 1 • simulated gate noise only

04 VQE VS QAOA

VQE • Estimates a low-energy state using a parameterized circuit and classical optimizer.

QAOA • Uses alternating cost and mixer layers to search for high-quality solutions to a combinatorial optimization problem.

PARAMETERIZED CIRCUIT + CLASSICAL OPTIMIZER | ALTERNATING COST + MIXER LAYERS

05 METHODOLOGY

1 DEFINE
Choose the Max-Cut problem

2 IMPLEMENT
Build VQE and QAOA circuits

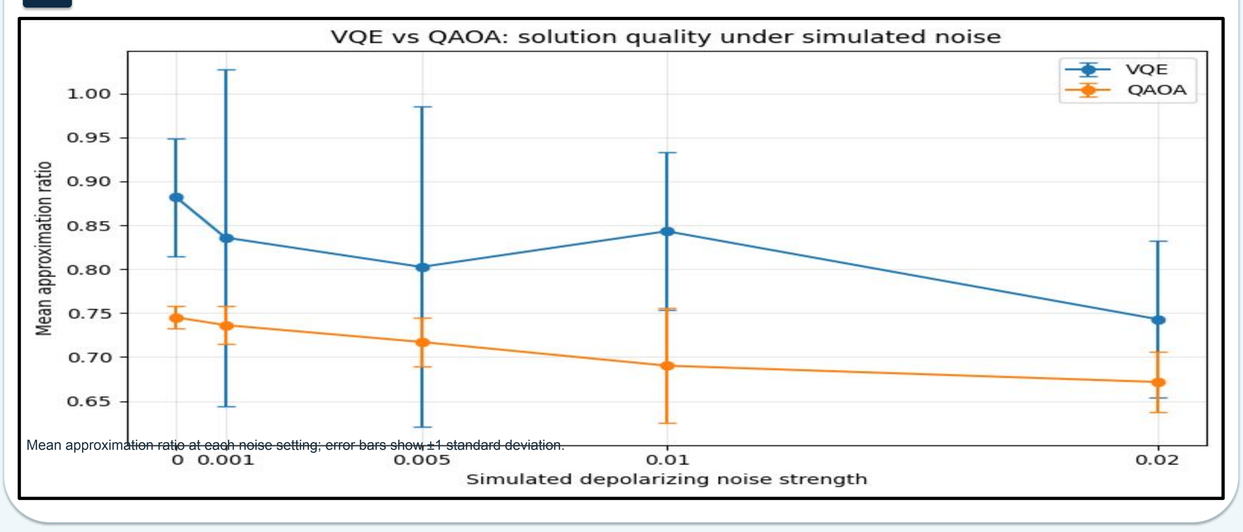
3 BASELINE
Run ideal simulations

4 ADD NOISE
Apply selected noise strengths

5 MEASURE
Collect objective values

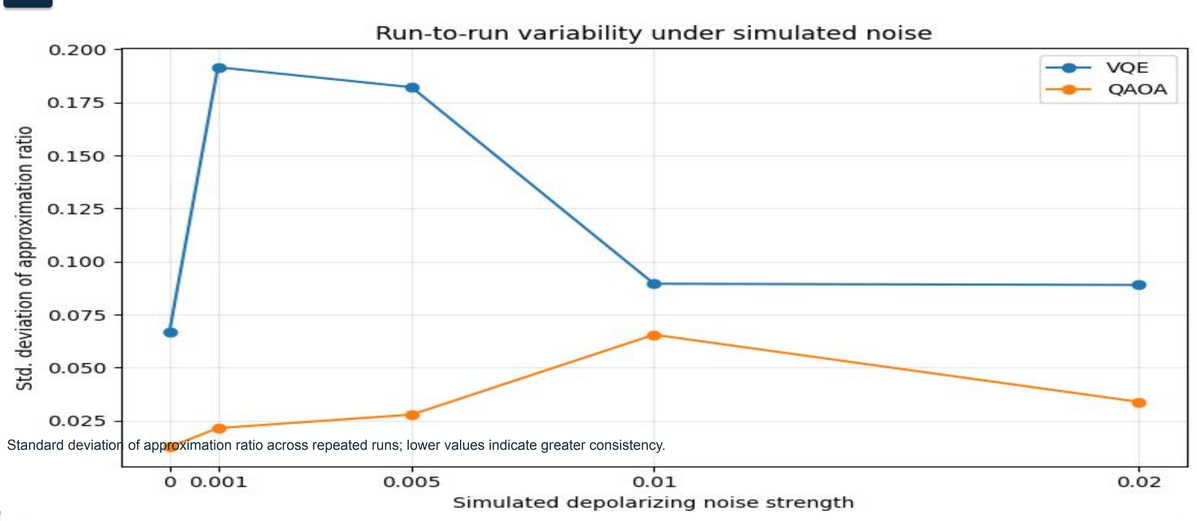
6 COMPARE
Compare quality and spread

06 SOLUTION QUALITY VS NOISE



Mean approximation ratio at each noise setting; error bars show ±1 standard deviation.

07 RUN-TO-RUN VARIABILITY



Standard deviation of approximation ratio across repeated runs; lower values indicate greater consistency.

08 KEY FINDINGS

- VQE has the higher mean approximation ratio at all five tested noise settings.
- Mean solution quality declines at the highest tested noise strength.
- QAOA has lower run-to-run standard deviation across the tested settings.
- Both algorithms show variability even in the ideal setting.

TAKEAWAY • Higher mean quality: VQE | Lower run-to-run spread: QAOA

09 INTERPRETATION

The observed results suggest a trade-off in this experiment: VQE achieves higher average solution quality, while QAOA is more consistent across repeated runs. The trends are not strictly monotonic, and should not be generalized to all problem sizes, circuit designs, or noise models.

INTERPRET WITHIN SCOPE • Four-node benchmark; simulated noise only

10 LIMITATIONS & FUTURE WORK

LIMITATIONS
- Simulated depolarizing noise may not capture every hardware effect.
- Results depend on problem size, circuit design, optimizer, and shot count.
- A four-node benchmark may not generalize to larger systems.

FUTURE WORK
- Validate on real quantum hardware.
- Test additional noise channels and circuit depths.
- Evaluate error-mitigation methods.

NEXT STEP • Validate trends on hardware and across noise models

11 CONCLUSION

For this four-node Max-Cut simulation, VQE produced higher mean approximation ratios, while QAOA showed lower run-to-run variability. These results indicate a performance–consistency trade-off under the tested simulated depolarizing-noise settings. They do not establish universal superiority or hardware robustness.



QUALITY vs CONSISTENCY — a measured trade-off

12 REFERENCES & REPRODUCIBILITY

1. Peruzzo, A. et al. (2014). "A variational eigenvalue solver on a photonic quantum processor." Nature, 512, 414–417.
2. Farhi, E., Goldstone, J., & Gutmann, S. (2014). "A Quantum Approximate Optimization Algorithm." arXiv:1411.4028.
3. Preskill, J. (2018). "Quantum Computing in the NISQ era and beyond." Quantum, 2, 79.

Reproducibility: Qiskit Aer simulation with selected depolarizing-noise strengths. Interpret findings only within the benchmark and settings described above. Verify bibliographic details against the sources used in the final study.

STUDY AT A GLANCE

4-NODE MAX-CUT	2 ALGORITHMS	SIMULATED NOISE
Benchmark problem	VQE and QAOA	Selected depolarizing strengths

Interpretation is limited to the stated benchmark and simulation settings; bibliographic details should be verified against the final study.